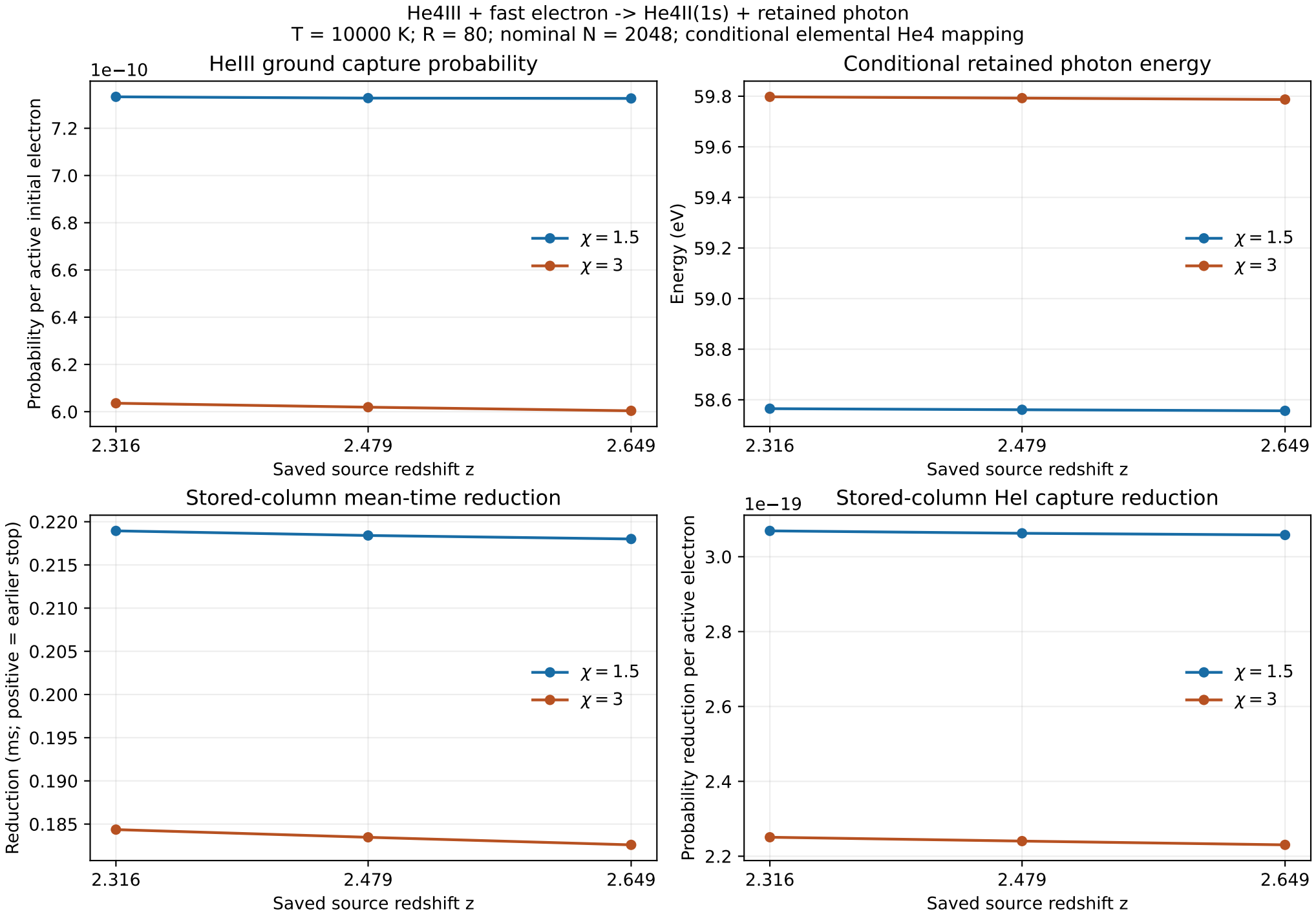
He4III + fast electron -> He4II(1s) + retained photon
T = 10000 K; R = 80; nominal N = 2048; conditional elemental He4 mapping
HeIII ground capture probability
1e−10
Probability per active initial electron
7.2
7.0
6.8
6.6
6.4
6.2
6.0
χ = 1.5
χ = 3
2.316
2.479
2.649
Saved source redshift z
Conditional retained photon energy
Energy (eV)
59.8
59.6
59.4
59.2
59.0
58.8
58.6
χ = 1.5
χ = 3
2.316
2.479
2.649
Saved source redshift z
Stored-column mean-time reduction
Reduction (ms; positive = earlier stop)
0.220
0.215
0.210
0.205
0.200
0.195
0.190
0.185
χ = 1.5
χ = 3
2.316
2.479
2.649
Saved source redshift z
Stored-column HeI capture reduction
1e−19
Probability reduction per active electron
3.0
2.8
2.6
2.4
2.2
χ = 1.5
χ = 3
2.316
2.479
2.649
Saved source redshift z